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Project Report

PORT SCANNER

Port Scanner Project

1. Project Overview

The Port Scanner is a basic cybersecurity project developed using Python. It is designed to check network ports on a target system and identify which ports are open. This project helps beginners understand network security and the basic working of port scanning.

2. Project Objectives

The main objectives of this project are:

- To understand the concept of network ports.
- To learn basic Python socket programming.
- To identify open ports on a target system.
- To understand the role of port scanning in cybersecurity.
- To gain practical experience in developing a basic cybersecurity tool.

3. Features

The project provides basic port scanning functionality.

- Scans target ports.
- Checks network connections.
- Displays scanning results in the terminal.
- Helps users understand open ports.
- Uses Python for simple implementation.

Note: The listed features should match the actual functionality of the source code.

4. Technologies Used

- **Python:** Used to develop the port scanner.
- **Socket Module:** Used to establish network connections and check port availability.
- **Visual Studio Code:** Used to write and edit the Python code.
- **Terminal:** Used to execute the program and display results.
- **GitHub:** Used to store and share the project source code.

5. Installation and Setup Instructions

Follow these steps to run the project:

1. Install Python on your computer.
2. Download or clone the project repository from GitHub.
3. Open the project folder in Visual Studio Code.
4. Open the terminal in Visual Studio Code.

5. Run the Python file using the following command:

```
python "port sccanner.py"
```

6. Follow the program's instructions and view the scanning results in the terminal.

Note: The command above uses the original filename. If the filename changes, update the command accordingly.

6. Usage

The Port Scanner can be used to check port availability on a target system for which the user has permission.

General steps:

1. Start the Python program.
2. Provide the required target information if requested by the program.
3. Allow the program to scan the configured ports.
4. Observe the results displayed in the terminal.
5. Review the results to understand which ports are open or unavailable.

The exact steps depend on the implementation of the Python source code.

7. System Workflow and Architecture

The general workflow of a port scanner is:

1. **Program Initialization:** The Python program starts execution.
2. **Target Selection:** The target IP address or hostname is provided.
3. **Port Selection:** The program identifies the ports to scan.
4. **Connection Attempt:** The socket module attempts to connect to the selected ports.
5. **Result Identification:** The program determines the connection status based on its implementation.
6. **Output Display:** The scanning results are displayed in the terminal.

Basic Workflow

Start Program → Select Target → Select Ports → Check Connections → Display Results

8. Testing and Results

Testing is important to verify that the program works as expected.

The following test cases can be used:

Test Case	Expected Result
Run the Python file	The program starts without errors
Scan an authorized target	The program attempts the configured connections
Check an open port	The program identifies a successful connection
Check a closed port	The program reports an unsuccessful connection
Provide invalid input, if supported	The program handles the input appropriately

Actual results should be recorded after performing these tests. Screenshots of the program execution and output can be added to this section.

9. Challenges Faced

During the development of this project, the following challenges may arise:

- Understanding the concept of network ports.
- Learning how Python socket programming works.
- Handling connection errors and timeouts.
- Understanding the results returned by network connections.
- Setting up and running the project in Visual Studio Code.

10. Future Improvements

The project can be improved in several ways:

- Add support for scanning multiple ports efficiently.
- Improve error handling and input validation.
- Save scanning results to a text or CSV file.
- Add a graphical user interface.
- Improve the presentation of scanning results.
- Add scanning progress information.

These improvements can make the tool easier to use and more informative.

11. Conclusion

The Port Scanner project provides an introduction to Python programming, network ports, and basic cybersecurity concepts. It demonstrates how socket connections can be used to check port availability. Developing this project helps build practical programming skills and provides a foundation for learning more advanced network security tools.

12. References

1. Python Official Documentation: https://docs.python.org/3/
2. Python Socket Programming Documentation: https://docs.python.org/3/library/socket.html
3. GitHub Documentation: https://docs.github.com/

13. Project Repository

GitHub Repository: Add your GitHub repository URL here.

https://github.com/eman123651/Port-Scanner-Project.git